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The effect of boundary-spanning search and cognitive style on the business model design of Internet start-ups: evidence from China

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ABSTRACT

Internet start-ups have the most dynamic organisational structure due to the rapid growth of network information technology. Based on social cognition theory and information processing theory, this study investigates the connection between boundary-spanning search and business model design, proposing two types of cognitive styles: adaptive and innovative. Based on the questionnaire survey data of 223 Internet start-ups, this study finds that: (1) boundary-spanning search of technical knowledge is positively related to the efficiency-centered business model. (2) boundary-spanning search of market knowledge is positively related to the novelty-centered business model; (3) hybrid boundary-spanning search is positively related to the combined business model. (4) cognitive style weakens or enhances the relationship between boundary-spanning search and business model design. The research provides practical guidance for Internet start-ups and contributes to a better understanding of the antecedent variables of the business model.

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1. Introduction

Business models, which are new systems of activities designed by firms for themselves and their stakeholders, play an important role in the business competition of Internet start-ups (Jin et al. 2022). Especially in the fierce trade war and technology war environment, business model design is crucial for the survival of enterprises and can help Internet start-ups overcome the innovation dilemma of broken chain and neck, enabling them to achieve rapid growth (Pedersen, Gwozdz, and Hvass 2018; Sohl, Vroom, and McCann 2020). Boundary-spanning search, as an important source of information for business model design (Yan et al. 2020; Yang, Wang, and Zhang 2021), provides Internet start-ups with access to external knowledge. This is primarily due to the rapid development in areas such as cloud computing and digital technologies, empowering Internet start-ups to leverage their strengths and rechange the connections between different organisations and business practices, improve the efficiency of their connections, and optimise resource allocation. However, not all Internet start-ups can effectively utilise these external resources for business model innovation. For example, Blockbuster acquired user rental data but failed to innovate and improve its business model, leading to its

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replacement by rivals like Netflix and other online streaming platforms (Teece 2010). Similarly, Kodak, once a dominant force in the film camera market, failed to adapt to externally acquired information and emerging trends in digital photography. This ultimately led to its downfall and bankruptcy in the digital photography industry. Undoubtedly, some Internet start-ups have accelerated the digitalisation process by engaging in boundary-spanning search, breaking information limitations, and leveraging it as a critical lever for business model innovation. However, some companies have failed to achieve business model innovation despite having access to external markets or technical knowledge. Therefore, it is valuable to explore the practical question of why some Internet start-ups can access diversified information and realise business model design through boundary-spanning search, while others are not.

Business model design is a process of analysis, planning, and innovation aimed at constructing frameworks and strategies to achieve business goals. Current research focuses on business model design from the perspectives of social networks, resource theory, and information processing theory. In summary, the social network perspective argues that mechanisms such as partnerships, information flow, and resource exchange within social networks contribute to the successful implementation of business models (Anand et al. 2021). The resource theory perspective emphasises the importance of organisational resource acquisition, integration, and innovation for business model design (Gerasymenko, De Clercq, and Sapienza 2015; Liu, Gu, and Liu 2022). The information processing theory perspective focuses on how to acquire information from the social environment, continuously learn, and adapt to market changes to optimise business model design and implementation, and has also gained attention (Martins, Rindova, and Greenbaum 2015). In business model design, boundary-spanning search is considered an important means to access heterogeneous resources, providing firms with knowledge of technology, markets, and more (Argote, Lee, and Park 2021; Yan et al. 2020). It plays a key influential role in business model design. However, there is no unanimous consensus on the research regarding boundary-spanning search. Some scholars argue that boundary-spanning search stimulates innovative activities in firms' products and services, accelerates business model changes, and achieves breakthrough innovations (Lopez-Vega, Tell, and Vanhaverbeke 2016; Yang, Wang, and Zhang 2021). Others point out that excessive boundary-spanning search may lead to knowledge redundancy, which can suppress firms' business model design (Terjesen and Patel 2017; Waeger and Weber 2019). From the above viewpoints, two problems can be identified: (1) There are conflicting views on the specific effects of boundary-spanning search on business model design, which raises doubts about its impact. (2) The newly acquired knowledge through boundary-spanning search is often scattered and disordered and may differ from the existing knowledge base. The organisation's ability to transform this knowledge into innovation capability and achieve breakthrough performance and improved market competitiveness depends on the way and ability of information processing.

Considering the existing limitations, this study argues that cognitive style plays a crucial role. This is because the value of information depends on how entrepreneurs process it, rather than the information itself. In social cognitive theory, the entrepreneur's information processing style is referred to as cognitive style. Different information processing styles lead to different outcomes (Boudreaux, Nikolaev, and Klein 2019). Scholars have identified various dimensions of cognitive styles, with Kirton (1976) and his theory of adaptation-innovation being widely recognised. When examining the relationship between boundary-spanning search and business model design, it is essential to consider the influence of individual cognitive styles on the underlying mechanisms. Based on information processing theory and social cognitive theory, this study collected data from 223 Internet start-ups through a questionnaire survey and constructed a theoretical model of boundary-spanning search on business model design. Additionally, the cognitive style was introduced as a moderating variable to effectively compensate for the shortcomings of existing studies.

The paper contributes primarily in the following areas. First, by establishing a connection between boundary-spanning search and business model design in Internet start-ups, enhancing the current literature on how such search strategies can benefit these start-ups. Considering the

inherent disadvantages faced by Internet start-ups, such as new entrant defects, our findings provide practical guidance for Internet start-ups to overcome resource constraints, gain competitive advantages, and address the challenges they encounter during their growth. Hence, these findings are crucial for the development of Internet start-ups. Second, this study demonstrates that the cognitive style of entrepreneurs plays a dual role in enhancing business model design through boundary-spanning search. Creative and open-minded entrepreneurs can derive inspiration from different industries, leading to unique and competitive business models. However, assimilationist entrepreneurs may overlook industry-specific factors, limiting the effectiveness of boundary-spanning search in generating innovative business models. Third, this study introduces social cognitive theory to refine the examination of business model design from a cognitive perspective. Entrepreneurs are influenced not only by individual cognitive factors but also by the social environment. Therefore, it is crucial to consider the impact of the social environment on entrepreneurs' cognition and the role of social cognitive factors in shaping business models. This approach complements and contributes to the cognitive research on business model design.

2. Literature review

The business model is the fundamental framework that enables enterprises to create and capture value (Amit and Zott 2015). In line with the 'antecedent-result' oriented business model research, entrepreneurs design business models by considering internal and external influencing factors, which can be divided into two types: the positioning perspective and the cognitive perspective.

2.1. Research from a positioning perspective

The positioning perspective, which has been widely adopted in theoretical and practical circles, focuses on understanding and designing business models. Scholars define a business model as a structure for value creation, emphasising the integration of its fundamental elements (Morris, Schindehutte, and Allen 2005). These elements can be combined in different ways to create various business models. Some scholars emphasise the modularity of business model elements, facilitating their standardised deconstruction and combination to understand the logical relationships involved in value creation. Morris, Schindehutte, and Allen (2005) argue that a business model comprises internal elements such as operational structure, economic logic, and strategic direction. Spieth and Schneider (2016) propose a three-dimensional framework consisting of value creation, value provision, and profitability. Other scholars adopt a factor valorisation perspective, focusing on aligning with the internal and external development environment and enhancing the value of the factors involved. Teece (2010) presents a circular logic model for business model design, considering environmental analysis, available technologies, customer benefits, target markets, revenue sources, and value capture mechanisms.

From the perspective of completely rational decision-making, entrepreneurs are believed to strive to maximise value. Research from the positioning perspective views business models as purposefully designed systems, with entrepreneurs portrayed as highly rational decision-makers.

2.2. Research from a cognitive perspective

The limited rationality theory complements the positioning perspective by recognising that entrepreneurs' decision-making processes are influenced by prior experiences and external contextual factors. Some scholars argue that entrepreneurs rely on satisfactory choices based on prior experience rather than optimising decisions based on economic considerations (Moqaddamerad and Tapinos 2023). The design of a new business model is influenced by different cognitive schemas, guiding a cognitive construction process (Islam 2019). Acquiring external knowledge is believed to compensate for inadequate resources and capability structures within enterprises, positively

influencing business model design (Lee and Huh 2016). Boundary-spanning search, which involves seeking external information and collaborating across organisational boundaries, contributes to the accumulation and renewal of organisational knowledge. It enables stakeholders to collaborate in the innovation process and broaden the range of innovation sources. Enterprises prioritise external resources and collaboration to enhance competitiveness. Boundary-spanning search also reduces path dependency, integrates external and internal knowledge, disrupts internal homogeneity, and drives transformative changes in business philosophy, processes, and organisational structure. These developments are driven by advancements in information technology and the increasing importance placed on external cooperation.

Although the positioning perspective and the cognitive perspective follow different decision-making logics, they are not isolated from each other. Instead, they mutually complement each other in research viewpoints.

3. Conceptual model and hypothesis development

3.1. *Boundary-spanning search of technical knowledge and efficiency-centered business model*

Utilising social environmental information enhances organisational innovation and flexibility. Technology introduction and R&D cooperation provide diverse technology-related knowledge, helping businesses overcome obstacles and knowledge gaps. This allows them to navigate the ‘familiarity trap’, evolve, and strengthen their foundation for an effective business model. By combining new knowledge with commercial value, enterprises enhance technological innovation efficiency and create value. For example, Duan et al. (2022) show that boundary-spanning search for technical knowledge positively affects innovation performance. It helps firms overcome technological lock-in by multinational companies, acquire external technological knowledge, reduce dependence, and enhance productivity. Lyu et al. (2022) found that technological boundary-spanning search enhances innovation performance by addressing technological inertia, fostering technical talents, and adopting advanced technologies for rapid corporate development.

Furthermore, industry leaders’ success contributes to organisational learnability and development ease. Leveraging existing technical knowledge reduces the risk of failure in utilising new knowledge, saving resources (Christofi et al. 2019). This approach is accessible and often the preferred choice for entrepreneurs. Lin and Li (2013) argue that a boundary-spanning search of technological knowledge enables companies to surpass the limitations of their existing technology and even address practical problems through radical means. This search unlocks possibilities for technological innovation that can potentially disrupt the entire industry. Therefore, we propose the following hypothesis:

H1: Boundary-spanning search of technological knowledge positively contributes to the efficiency-centered business model.

3.2. *Boundary-spanning search of market knowledge and novelty-centered business model*

Boundary-spanning search of market knowledge facilitates organisations in identifying untapped customer segments, enabling the establishment of novel business models (Sidhu, Commandeur, and Volberda 2007). This search breaks cognitive patterns and behaviours by accessing new market information and resources, driving new product development and effective market response (Lin and Li 2013; Yang, Wang, and Zhang 2021). Integration of cutting-edge external knowledge into innovative marketing channels enhances organisational performance, creating new distribution channels and improving profitability. Zhang et al. (2017) found that demand-side search improves

firms' understanding of product positioning, market demand, and customer preferences, leading to enhanced business model innovation and customer value proposition fulfilment.

Transferring information over time leads to applied innovation. Business model innovation encompasses two key aspects: restructuring and reorganisation of internal governance structures and governance elements, as well as establishing value propositions that transcend organisational boundaries and foster win-win cooperation with stakeholders (Zott and Amit 2008). Firms pursue new and diverse knowledge, intensifying market searches across borders to absorb external opportunities and generate sustained momentum for innovative business models. Lopez-Vega, Tell, and Vanhaverbeke (2016) contend that boundary-spanning search enables the incorporation of rich heterogeneous information into the organisation's knowledge base. It facilitates the assessment of the external competitive environment and prompts an adjustment of the organisation's market exploration behaviour and has a profound impact on promoting product renewal. Therefore, we propose the following hypothesis:

H2: Boundary-spanning search of market knowledge positively contributes to the novelty-centered business model.

3.3. Hybrid boundary-spanning and combined business model

The previous section suggests that boundary-spanning searches for technical knowledge and market knowledge search have positive effects on the efficiency and novelty of business models, respectively. However, further research is needed to determine the potential synergies of combining the two in a combined business model for companies. Wang and Chung (2020) argue that businesses engaged in boundary-spanning searches must carefully consider various market environments and corporate management factors to quickly adapt their action guidelines and effectively integrate organisational resources. Therefore, both boundary-spanning searches for market knowledge and technical knowledge are interdependent and essential for designing a robust business model. Businesses engaged in technical knowledge boundary-spanning search can, on one hand, benefit from the outcomes of market knowledge boundary-spanning search. As per Cho and Kim (2017)'s study, a successful technical knowledge boundary-spanning search provides strategic direction for market knowledge boundary-spanning search, enhancing the ability to predict and understand external market trends. This streamlined process of information screening and customer selection contributes to more innovative performance by businesses. Alternatively, a boundary-spanning search for technological knowledge can benefit from a boundary-spanning search for market knowledge. By integrating technical and market information with internal development strategies, Enterprises can enhance their sustainable competitive advantage. Given the impact of Internet technology, cross-border integration has become a vital strategy for synergizing business models.

A hybrid boundary-spanning search, which connects technical knowledge and market knowledge, has the potential to integrate and reconfigure an organisation's resources in response to changing market demands. This integration enables the creation of business models with disruptive qualities. Therefore, we propose the following hypothesis:

H3: Hybrid boundary-spanning search has a positive contribution to the combined business model.

3.4. The moderating effect of cognitive style

3.4.1. The moderating role of adaptive cognitive style

Assimilationist behavioural strategies are commonly favoured by individuals with an adaptive cognitive style, as they tend to rely on previous systems and methods to solve problems. However, leveraging the connections between internal and external information provides a practical means of escaping the familiarity trap. Boundary-spanning search for technological knowledge involves

gathering information related to production, design, and other aspects. When business owners adopt an adaptive cognitive style, they often rely heavily on existing models and procedures, being less inclined to question their viability. This inclination can lead them to dismiss external knowledge as unreliable information, thereby hindering their ability to improve the accuracy and efficiency of their work and impeding the development of effective business models. According to Lomberg, Kollmann, and Stöckmann (2017), individuals with adaptive cognitive styles prefer to solve problems incrementally, making small adjustments based on prior knowledge. However, this approach can be counterproductive to increasing a company's operational efficiency.

Boundary-spanning search for market knowledge can provide business owners with valuable market and customer information that can be sorted and utilised. However, when business owners adopt an adaptive cognitive style, they often approach problems conservatively and cautiously, overlooking important details. This can limit the creative output of the business. According to von Wittich and Antonakis (2011), individuals with adaptive cognitive styles are constrained by existing paradigms and external rules, limiting the depth and flexibility of their thinking. Furthermore, Baluku and Otto (2019) found that adaptive cognitive styles prioritise regularity and may impede the development of creative thinking. Consequently, entrepreneurs with adaptive cognitive styles may focus on enhancing performance within the current framework but struggle to adapt to environmental changes and competitive mechanisms. Therefore, we propose the following hypothesis:

H1a: Adaptive cognitive style negatively moderates the relationship between the boundary-spanning search of technological knowledge and efficiency-centered business model.

H2a: Adaptive cognitive style negatively moderates the relationship between the boundary-spanning search of market knowledge and novelty-centered business model.

H3a: Adaptive cognitive style negatively moderates the relationship between hybrid boundary-spanning search and combined business model.

3.4.2. The moderating role of innovative cognitive style

Individuals with an innovative cognitive style are driven to develop new problem-solving approaches and prioritise the originality of solutions. They recognise the importance of thinking outside the box to generate fresh perspectives. Boundary-spanning activities enable managers to access external resources, expanding the information pool for innovation. Transforming new knowledge into organisational innovation and competitive advantage is crucial. Entrepreneurs with an innovative cognitive style are more likely to adopt new technology for increased productivity and profitability. Baluku and Otto (2019) found that individuals with innovative cognitive styles excel at extracting useful information from interpersonal interactions and place significant emphasis on learning from others' experiences. These characteristics have a positive impact on employee innovation self-efficacy.

Innovative business owners adopt an adventurous and open problem-solving approach, driving ground-breaking innovations. Assimilating and integrating valuable knowledge contributes to innovation performance and creates value. Managers with an innovative cognitive style integrate external knowledge, cultivate creative thinking, and explore ideas from the external world. Carnabuci and Diószegi (2015) find a positive correlation between creative style and innovation behaviour in organisations. Individuals with a creative style seek to transform organisational structures and systems, deriving satisfaction from generating fresh solutions. Entrepreneurs with an innovative cognitive style explore alternative opportunities, address new challenges, and embrace novel ventures, surpassing the current paradigm. Therefore, we propose the following hypothesis:

H1b: Innovative cognitive style positively moderates the relationship between the boundary-spanning search of technological knowledge and efficiency-centered business model.

H2b: Innovative cognitive style positively moderates the relationship between the boundary-spanning search of market knowledge and novelty-centered business model.

H3b: Innovative cognitive style positively moderates the relationship between hybrid boundary-spanning search and combined business model.

The above analysis was combined to construct the research model shown in [Figure 1](#).

4. Research methodology

4.1. Survey instrument design

Business model design, boundary-spanning search, and cognitive style are the three key variables in this study. Each variable is assessed using a five-point Likert scale.

4.1.1. Business model design

The business model scale developed by Zott and Amit (2007) is utilised to measure the efficiency-centered business model (EBM) (7 items) and novelty-centered business model (NBM) (9 items). The combined business model (CBM) is mainly measured by the product term of efficient business models and novel business models (Hu and Chen 2016).

4.1.2. Boundary-spanning search

The boundary-spanning search scale is adapted from Sidhu, Commandeur, and Volberda (2007) and comprises two dimensions: boundary-spanning search of technical knowledge (BSTK) (6 items) and boundary-spanning search of market knowledge (BSMK) (6 items). Hybrid boundary-spanning search (HBS) is measured by constructing the product term of technical knowledge boundary-spanning search and market knowledge boundary-spanning search.

4.1.3. Cognitive style

The adaptive cognitive style (ACS) consists of eight items, while the innovative cognitive style (ICS) consists of five items. Both cognitive styles are derived from Kirton (1976)’s ‘adaptive-innovative’ theory and its measurement.

4.1.4. Control variables

Drawing on previous research on business model design (Liu, Gu, and Liu 2022), boundary-spanning search (Zhu, Miao, and Jin 2022), and cognitive style (Baluku and Otto 2019), this study considers control variables that account for both firm-level and individual-level effects. At the firm level, the study concluded that firm age, industry field, and asset size affect the business model design

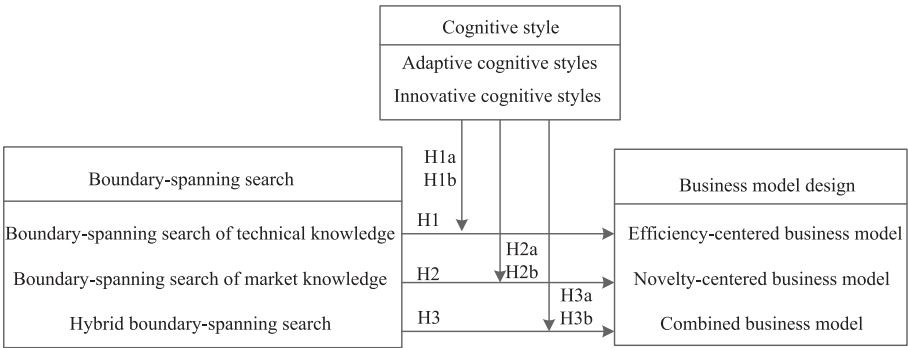


Figure 1. Theoretical framework.

process. For example, larger firms tend to have more resources available, leading to a propensity for designing highly innovative business models. Consequently, the size of assets at the time of firm creation is controlled (Kim and Min 2015). Additionally, individual-level variables such as entrepreneurial gender, age, and educational background are also taken into account. To summarise, control variables in this study included firm age, industry field, asset size, entrepreneur's gender, age, and educational background.

4.2. Sample and data collection

This paper focuses on entrepreneurial enterprises in the Internet industry. This choice is based on the research conducted by Zott and Amit (2007) and Metallo et al. (2018), which primarily explores business model design within the Internet industry. The Internet industry is characterised by rapid and continuous development, constantly introducing new models and business concepts. It demonstrates higher adaptability and dynamism, making it conducive to the exploration of innovative business models. Consequently, this research is limited to entrepreneurial enterprises within the Internet industry. Following Zahra (2021)'s perspective, start-ups with an establishment year of fewer than 8 years are selected as research samples, using the establishment year as the criterion to define start-ups.

Data collection occurred between June 2021 and September 2021, targeting entrepreneurs and core members of start-up teams. The questionnaires were administered through the data research website Wenjuanxing and among MBA members of the school. A total of 428 questionnaires were distributed, and 334 were successfully collected, yielding a recovery rate of 78.04%. After excluding questionnaires that did not meet the criteria, such as incomplete or erroneous responses and failure to meet the establishment year criteria, 223 valid questionnaires remained, resulting in an effective recovery rate of 66.77%. Descriptive statistics for the sample data are presented in Table 1.

4.3. Reliability and validity analysis

Table 2 shows satisfactory internal consistency for each variable, with minimum Cronbach's alpha coefficient of 0.767 (>0.7 recommended). Additionally, the CR combined reliability coefficient ranged from 0.844 to 0.993, indicating good internal consistency. Validity was assessed using factor loadings and AVE. The factor loadings of the measured question items for each relevant variable were greater than 0.6, and the AVE values for each latent variable were greater than 0.5, indicating good convergent validity. The results in Table 3 indicate that the square root of the AVE for each variable was greater than the correlation coefficient, thereby confirming good discriminant validity.

Using AMOS 24.0 software, the baseline model (six-factor model) had a χ^2/df value of 1.071, an RMSEA value of 0.018, a CFI value of 0.981, an IFI value of 0.982, and a TLI value of 0.980. The baseline model (six-factor model) had the best fit compared to the other measurement models, suggesting that the six research variables in this study have high discriminant validity and represent the six different constructs.

4.4. Common method bias

To control for common method bias, several measures were implemented ex-ante. Firstly, the questionnaire included multiple objective variables like firm age. Additionally, various question formats, such as single-choice and Likert scales, were utilised. Secondly, reversal questions were incorporated to counteract automatic responses, allowing for careful evaluation. Third, the guidance section explicitly stated that the research was for academic purposes, emphasising no right or wrong answers.

Further, the common method bias was tested by Harman's single-factor test. The results showed that KMO was 0.859, the approximate Chi-square was 3489.878, the degree of freedom was 820, and

Table 1. Sample characteristics.

Characteristics	Number	Proportion
Firm age		
≤1	14	6.3
2	52	23.3
3	63	28.3
4	21	9.4
5	17	7.6
6	21	9.4
7	20	9
8	15	6.7
Industry field		
Basic application	72	32.2
Business transaction	52	23.3
Network finance	16	7.2
Network entertainment	38	17
Public service	26	11.7
Others	19	8.5
Assets size (million)		
<0.5	35	15.7
0.5–0.99	65	29.1
1–4.99	90	40.4
5–9.99	22	9.9
>10	11	4.9
Gender		
Male	126	56.5
Female	97	43.5
Entrepreneur age		
<25	19	8.5
26–30	91	40.8
31–35	80	35.9
36–40	18	8.1
>41	15	6.7
Education		
High school/junior college and below	8	3.6
Junior college education	26	11.7
College	140	62.8
Master	39	17.5
PhD	10	4.5

the significance level was 0.000. The explanatory power of the first common factor extracted in the non-rotating state was 20.66%, less than the standard of 50%. It indicates that the common method bias is within the acceptable range.

5. Results

5.1. Correlation analysis

Table 3 presents the correlations, means, and standard deviations of the key constructs. The VIF values for each variable were below 2, indicating no multicollinearity in the data, which wouldn't affect the subsequent analysis.

Table 2. Reliability and validity.

Variables	Loading	Cronbach's α	AVE	CR
BSTK	0.706–0.731	0.812	0.516	0.865
BSMK	0.701–0.715	0.800	0.501	0.858
ACS	0.696–0.728	0.858	0.503	0.890
ICS	0.704–0.767	0.767	0.520	0.844
EBM	0.709–0.759	0.851	0.530	0.887
NBM	0.694–0.723	0.875	0.503	0.901

Table 3. Correlation matrix and descriptive statistics.

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Firm age	1													
Industry field	−0.003	1												
Assets size	0.153*	0.003	1											
Gender	−0.054	−0.034	−0.023	1										
Entrepreneur age	−0.018	0.025	0.178**	0.357**	1									
Education	−0.138*	−0.011	−0.040	0.039	0.206**	1								
BSTK	0.003	−0.044	0.021	−0.099	−0.057	0.021	0.718							
BSMK	−0.127	−0.062	0.090	−0.047	0.079	0.136*	0.420**	0.708						
HBS	0.076	0.036	−0.042	−0.026	0.019	−0.019	0.213**	0.341**	1					
ACS	−0.001	0.030	0.004	−0.034	−0.190**	0.002	0.068	0.044	0.058	0.709				
ICS	−0.091	−0.039	0.043	−0.060	0.060	0.035	0.214**	0.336**	0.346**	0.037	0.721			
EBM	−0.118	−0.077	0.098	−0.061	0.033	0.048	0.431**	0.340**	0.461**	0.208**	0.232**	0.728		
NBM	−0.112	−0.103	0.122	−0.059	0.101	0.067	0.287**	0.401**	0.422**	0.113	0.425**	0.359**	0.709	
CBM	−0.148*	−0.012	0.158*	−0.159*	0.119	0.150*	−0.013	0.162*	0.183**	0.094	0.287**	0.164*	0.414**	1
Mean	3.865	2.780	2.592	1.565	2.637	3.076	3.596	3.509	0.862	3.632	3.872	3.861	3.934	0.793
SD	2.029	1.701	1.026	0.497	0.985	0.782	0.463	0.445	1.362	0.430	0.460	0.521	0.433	1.242

Note: $N = 223$; * $p < 0.05$; ** $p < 0.01$. Bold values are the square root of AVE.

5.2. Results of hypotheses testing

Regression models were utilised to examine the impact of boundary-spanning search on business model design in Internet start-ups (Table 4). M1 serves as the benchmark analysis, incorporating only the control variables. M2 builds upon M1 by including the effect of technological knowledge boundary-spanning search on the efficiency-centered business model. The results demonstrate a positive effect of technological knowledge boundary-spanning search on the efficiency-centered business model ($\beta = 0.478, p < 0.001$), supporting H1. M3 represents the benchmark analysis, including only the control variables. M4 extends M3 by incorporating the effect of a boundary-spanning search for market knowledge on the novelty-centered business model. The findings indicate a positive effect of a boundary-spanning search for market knowledge on the novelty-centered business model ($\beta = 0.358, p < 0.001$), supporting H2. M5 serves as the benchmark analysis, including only the control variables. M6, based on M5, includes the effect of hybrid boundary-spanning search on the combined business model. The results demonstrate a positive effect of hybrid boundary-spanning search on the combined business model ($\beta = 0.175, p < 0.001$), supporting H3.

Table 5 presents the results of the regression analysis examining the moderating effect of cognitive style. M7 incorporates the moderating variable adaptive cognitive style on top of M2. The findings reveal that the interaction term between boundary-spanning search for technical knowledge and adaptive cognitive style is negative and statistically significant ($\beta = -0.432, p < 0.05$), supporting H1a. M9, based on M4 with the moderating variable adaptive cognitive style, indicates that the interaction term between market knowledge boundary-spanning search and adaptive cognitive style is negative but not significant ($\beta = -0.248$), suggesting no significant interaction effect and failing to support H2a. M11, built upon M6 with the moderating variable adaptive cognitive style, demonstrates that the interaction term between hybrid boundary-spanning search and adaptive cognitive style is negative and statistically significant ($\beta = -0.334, p < 0.05$), supporting H3a.

Based on M2, M8 is a regression analysis incorporating the moderator variable innovative cognitive style. The results indicate that the interaction term between technological knowledge boundary-spanning searches and adaptive cognitive style is positive and statistically significant ($\beta = 0.268, p < 0.05$), supporting H1b. M4 serves as the basis for M10, which includes the moderating

Table 4. Regression model and results.

Variables	EBM		NBM		CBM	
	M1	M2	M3	M4	M5	M6
Firm age	-0.035* (-1.98)	-0.034* (-2.17)	-0.028 (-1.93)	-0.018 (-1.30)	-0.003 (-0.40)	-0.005 (-0.63)
Industry field	-0.025 (-1.21)	-0.019 (-1.02)	-0.028 (-1.66)	-0.022 (-1.38)	0.002 (0.24)	0.014 (1.12)
Assets size	-0.057 (1.62)	0.051 (1.61)	0.051 (1.77)	0.035 (1.31)	-0.038* (-2.37)	-0.033* (-2.23)
Gender	-0.086 (-1.14)	-0.046 (-0.68)	-0.094 (-1.52)	-0.068 (-1.17)	0.055 (1.72)	0.058 (1.85)
Entrepreneur age	0.019 (0.48)	0.027 (0.75)	0.049 (1.51)	0.038 (1.26)	0.062 (1.08)	0.016 (0.97)
Education	0.019 (0.42)	0.010 (0.25)	0.019 (0.49)	-0.004 (-0.11)	0.124 (1.70)	0.035 (0.07)
BSTK		0.478*** (7.00)				
BSMK				0.358*** (5.83)		
HBS						0.175*** (4.83)
R ²	0.041	0.219	0.060	0.189	0.074	0.186
Adj-R ²	0.014	0.193	0.034	0.162	0.048	0.172
F	1.521	8.603***	2.318*	7.140***	2.889*	3.692***

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 5. Results of a test of the moderating effect of cognitive style.

Variables	EBM		NBM		CBM	
	M7	M8	M9	M10	M11	M12
Firm age	−0.034* (−2.24)	−0.027 (−1.73)	−0.016 (−1.23)	−0.012 (−0.97)	−0.005 (−0.67)	−0.004 (−0.48)
Industry field	−0.024 (−1.33)	−0.011 (−0.60)	−0.020 (−1.30)	−0.020 (−1.37)	−0.001 (−0.13)	−0.003 (−0.32)
Assets size	0.040 (1.28)	0.048 (1.54)	0.031 (1.17)	0.032 (1.28)	0.032* (2.18)	0.053* (2.29)
Gender	−0.056 (−0.85)	−0.033 (−0.49)	−0.076 (−1.33)	−0.044 (−0.81)	−0.058 (−1.84)	−0.062* (−2.30)
Entrepreneur age	0.059 (1.67)	0.012 (0.34)	0.062* (1.97)	0.022 (0.76)	0.014 (0.82)	0.010 (0.62)
Education	0.007 (0.16)	0.011 (0.28)	−0.010 (−0.28)	0.006 (0.18)	−0.035 (−0.30)	0.035 (0.33)
BSTK	0.435*** (6.44)	0.453*** (6.57)				
BSMK			0.335*** (5.46)	0.251*** (4.15)		
HBS					0.184*** (5.53)	0.241*** (3.50)
ACS	0.232** (3.19)		0.114 (1.81)		0.236** (2.36)	
ICS		0.143* (2.04)		0.311*** (5.36)		0.250*** (3.49)
BSTK × ACS	−0.432* (−2.34)					
BSMK × ACS			−0.248 (−1.81)			
HBS × ACS					−0.334* (−2.36)	
BSTK × ICS		0.268* (2.02)				
BSMK × ICS				0.259** (2.46)		
HBS × ICS						0.233* (1.99)
R^2	0.275	0.246	0.214	0.296	0.230	0.298
Adj- R^2	0.244	0.214	0.181	0.267	0.224	0.253
F	8.963***	7.708***	6.447***	9.963***	9.785***	8.592***

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

variable of innovative cognitive style. The findings demonstrate that the interaction term between boundary-spanning search for market knowledge and innovative cognitive style is positive and highly significant ($\beta = 0.259$, $p < 0.01$), supporting H2b. Building upon M6, M12 incorporates the moderator variable innovative cognitive style. The results reveal that the interaction term between hybrid boundary-spanning search and innovative cognitive style is positive and significant ($\beta = 0.233$, $p < 0.05$), supporting H3b.

6. Discussion and conclusions

6.1. Conclusion

The main findings of the study are as follows: First, technological knowledge boundary-spanning search significantly and positively influences the efficiency-based business models of Internet start-ups. Second, market knowledge boundary-spanning search can effectively enhance the level of novel business models of Internet start-ups. Third, hybrid boundary-spanning search has a positive impact on the combined business models of Internet start-ups. Fourth, cognitive styles influence the relationship between boundary-spanning search and business model design. Specifically, the

adaptive cognitive style has a significant negative effect on the execution effect of technical knowledge boundary-spanning search and hybrid boundary-spanning search, respectively, and does not have a significant effect on the execution effect of market knowledge boundary-spanning search. Innovative cognitive styles have a significant positive impact on the execution of boundary-spanning search for technical knowledge, boundary-spanning search for market knowledge, and hybrid boundary-spanning search, respectively.

6.2. Theoretical contributions

The theoretical implications of this study are threefold. Firstly, it emphasises that research from a cognitive perspective should not solely focus on meaning construction but also recognise the contribution of boundary-spanning search as another important cognitive perspective in the field of entrepreneurship, thereby enriching social cognitive theory. This study specifically examines the antecedents of boundary-spanning search, which aids in acquiring diverse knowledge and integrating it into organisational systems and practices to develop proprietary tacit knowledge with organisational characteristics. This, in turn, contributes to the establishment of sustainable competitive advantage (Wang, Xue, and Yang 2022). This finding aligns with Yang, Wang, and Yang (2021) view that boundary-spanning search not only provides organisations with diverse knowledge resources but also drives them to update their knowledge bases, enhance the efficiency of knowledge use, and explore new products and services that meet market needs.

Second, this paper focuses on the antecedent research of business model design and explores the differential mechanisms of various types of boundary-spanning searches on business model design. It emphasises the combination of dynamic environmental information acquired through boundary-spanning search and the enterprises' existing knowledge. Business model design is a cognitive process that involves the interaction between external knowledge and internal structure. By examining its antecedents from a content perspective, this research contributes to uncovering the scientific principles underlying the development of business model attributes. Existing studies on business models mostly concentrate on exploring the effects of business model outcomes on firm performance (Saebi, Lien, and Foss 2017). This study specifically explores the boundary-spanning search types that drive business model design. This enriches the understanding of the triggers for business model design and establishes a crucial mechanism connecting search and design.

Thirdly, this study breaks away from the existing research on the direct effect of 'boundary-spanning search-business model design' and adds to the understanding of entrepreneurs' cognitive traits by analysing the influence of cognitive style as a contextual variable on business model design. This adds to the cognitive school of research in the field of business model design. While previous research has not specifically focused on whether cognitive style affects firm development, this study explores the moderating role of the 'adaptive-innovative' cognitive style to understand the complex effects (Kozhevnikov, Ho, and Koh 2022). By investigating the impact of boundary-spanning search on business model design and the circumstances under which it occurs, this study addresses the overemphasis on single factors and enriches the theoretical explanation of business model design.

6.3. Practical implications

This study offers several practical implications. First, Internet start-ups should foster a learning culture, establish an external knowledge monitoring system, enhance information absorption and utilisation efficiency, and leverage boundary-spanning search for innovative solutions. Boundary-spanning search has evolved into a pathway for gaining competitive advantages. Technologies like AI and machine learning offer smarter tools for boundary-spanning search, automating data collection, organisation, and analysis. This enables more accurate and personalised search results and recommendations for enterprises (Wang, Xue, and Yang 2022). Second, entrepreneurs' cognitive

style significantly impacts the effectiveness of boundary-spanning searches. Access to diverse information is important, but integration and absorption of this information are vital for business model design. To cultivate innovative cognitive styles and address competition, enterprises should foster a culture of innovation, openness, and knowledge sharing. Encouraging managers to adopt an innovative mindset, embrace change, challenge traditional thinking, and explore new possibilities is crucial. Third, enterprises must enhance their business model design to adapt to evolving market needs. A well-operated business model can greatly improve business performance. In an uncertain environment, entrepreneurs should prioritise building unique business models that enable sustainable development in the Internet era. Internet start-ups should actively explore the integration of physical business elements and strive to create distinctive business models that differentiate them from competitors.

6.4. Research limitations and future directions

This study has some shortcomings that require further improvement. Firstly, the questionnaires were distributed primarily through Questionnaire Star, which may lead to measurement errors and impact data accuracy. Secondly, while the sample size meets the requirements, it could be larger. Future research should focus on the following improvement. Firstly, measuring key variables, such as boundary-spanning search and business models using a combined model. Analysing the constructed indicators for each variable in the combined model will help enrich the indicator pool and enhance the measurement accuracy. Secondly, integrating infrastructure attributes and value logic from strategic management and marketing fields can be explored to develop measurement scales and identify feasible directions for business model development.

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